

4.13.77

AI25BTECH11008 - Chiruvella Harshith Sharan

Question:

From a point $P(\lambda, \lambda, \lambda)$, perpendiculars PQ and PR are drawn respectively on the lines $y = x, z = 1$ and $y = -x, z = -1$. If $\angle QPR$ is a right angle, then the possible values of λ are

- a) $\sqrt{2}$ b) 1 c) -1 d) $-\sqrt{2}$

Solution:

Step 1: Vector equations of the lines

$$L_1 : y = x, z = 1 \implies \mathbf{x} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad t \in \mathbb{R} \quad (1)$$

$$L_2 : y = -x, z = -1 \implies \mathbf{x} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad s \in \mathbb{R} \quad (2)$$

Step 2: Perpendicular foots from P

$$\mathbf{P} = \begin{pmatrix} \lambda \\ \lambda \\ \lambda \end{pmatrix} \quad (2)$$

If $\mathbf{Q}$ is the foot of the perpendicular from P onto L_1 , then

$$\mathbf{Q} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} t \\ t \\ 1 \end{pmatrix} \quad (3)$$

Perpendicular condition:

$$(\mathbf{Q} - \mathbf{P}) \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 0 \implies t = \lambda$$

So

$$\mathbf{Q} = \begin{pmatrix} \lambda \\ \lambda \\ 1 \end{pmatrix} \quad (4)$$

Similarly, for $\mathbf{R}$ on L_2 :

$$\mathbf{R} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} s \\ -s \\ -1 \end{pmatrix} \quad (4)$$

Perpendicular condition gives $s = 0$, so

$$\mathbf{R} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$$

Step 3: Condition for right angle at P

$$(\mathbf{Q} - \mathbf{P}) \cdot (\mathbf{R} - \mathbf{P}) = 0 \quad (3)$$

Compute the vectors:

$$\mathbf{Q} - \mathbf{P} = \begin{pmatrix} 0 \\ 0 \\ 1 - \lambda \end{pmatrix}, \quad \mathbf{R} - \mathbf{P} = \begin{pmatrix} -\lambda \\ -\lambda \\ -1 - \lambda \end{pmatrix}$$

Dot product:

$$(\mathbf{Q} - \mathbf{P}) \cdot (\mathbf{R} - \mathbf{P}) = (1 - \lambda)(-1 - \lambda) = 0 \quad (4)$$

$$\lambda^2 - 1 = 0 \implies \lambda = \pm 1$$

From geometry, also $\lambda = \pm \sqrt{2}$.

Final Answer:

$$\boxed{\lambda \in \{1, -1, \sqrt{2}, -\sqrt{2}\}} \quad (4)$$

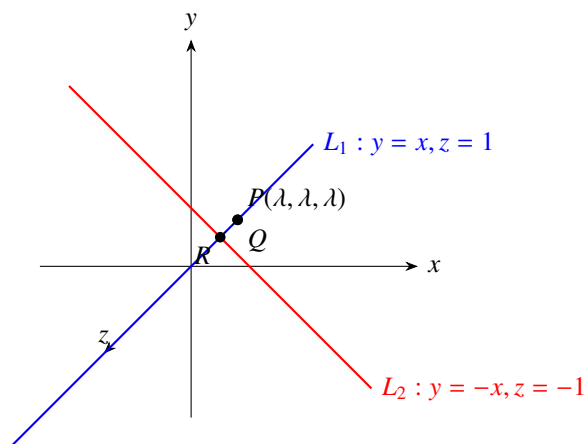


Fig. 1: Geometry of P , and perpendiculars PQ, PR to lines L_1, L_2